MT. TAPULAO · ZAMBALES MOUNTAIN RANGE · LUZON, PHILIPPINES

The mountain beneath

How does life change with altitude?

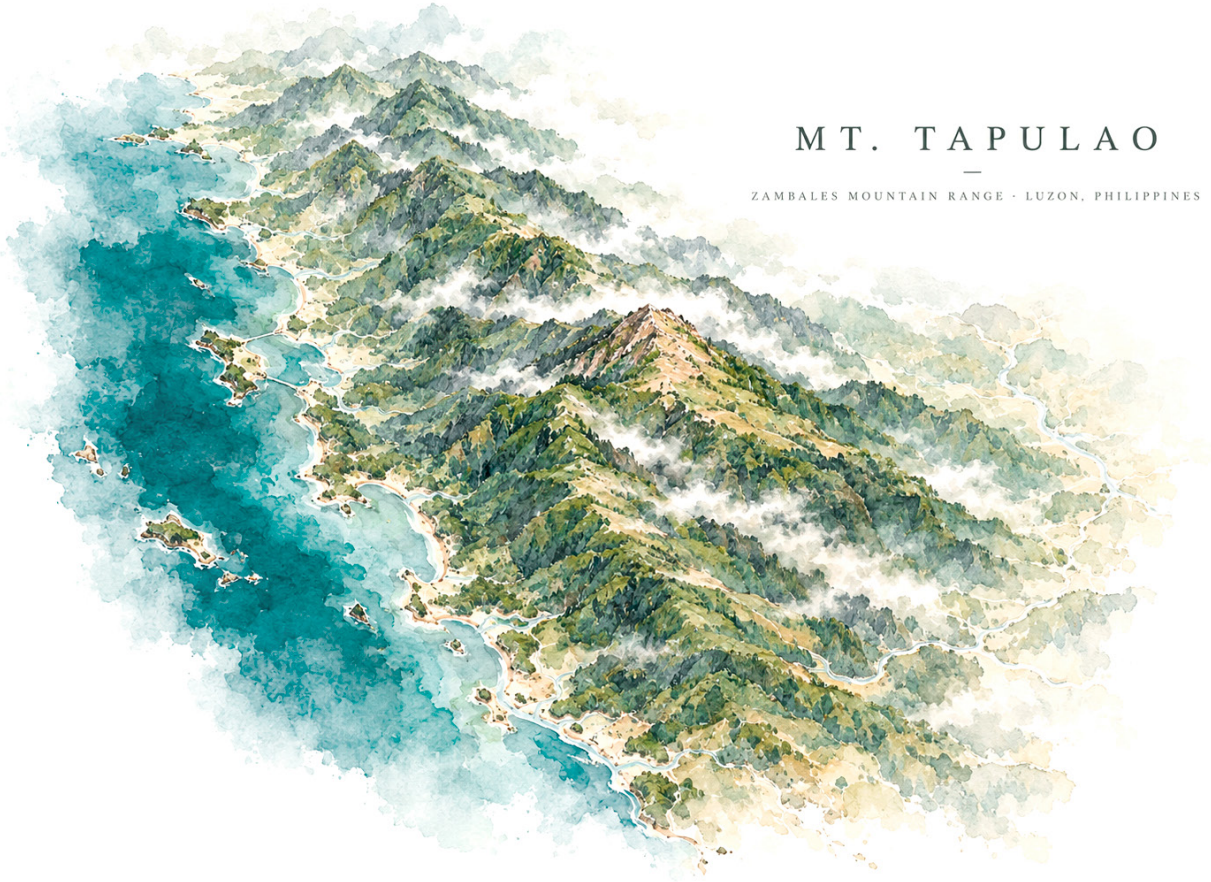
Five forest zones, hidden fungal worlds, and the unseen biodiversity of Mt. Tapulao.



A VISUAL SCIENCE STORY

Where this story begins

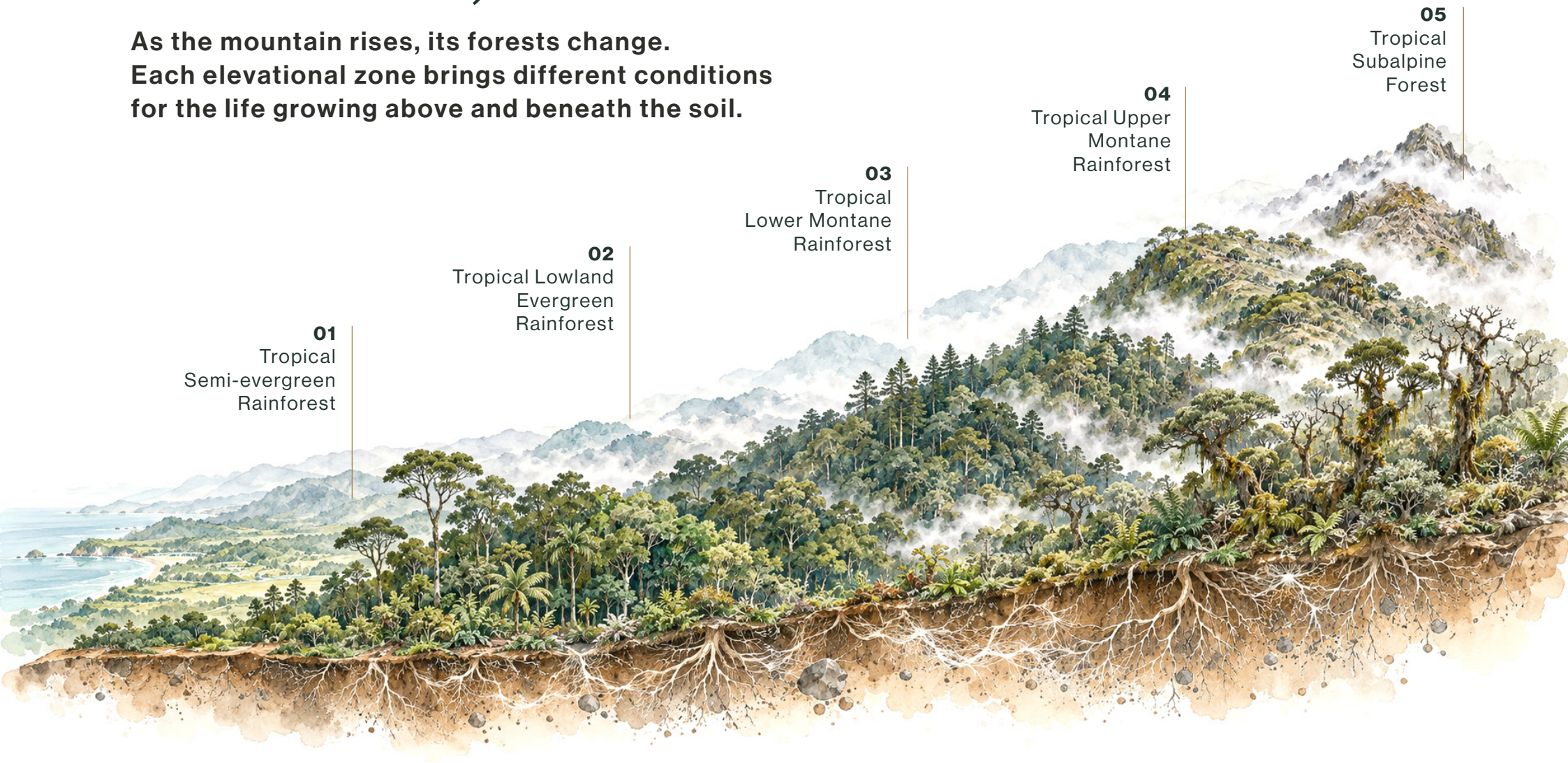
Mt. Tapulao is a mountain ecosystem in western Luzon, where altitude shapes forests, biodiversity, and the hidden life beneath the soil.



Here, five forest zones unfold along the slopes of the Zambales Mountain Range. As elevation changes, plants, animals, and fungal communities may shift with them, revealing an unseen mountain world still waiting to be understood.

Five forests, one mountain

**As the mountain rises, its forests change.
Each elevational zone brings different conditions
for the life growing above and beneath the soil.**

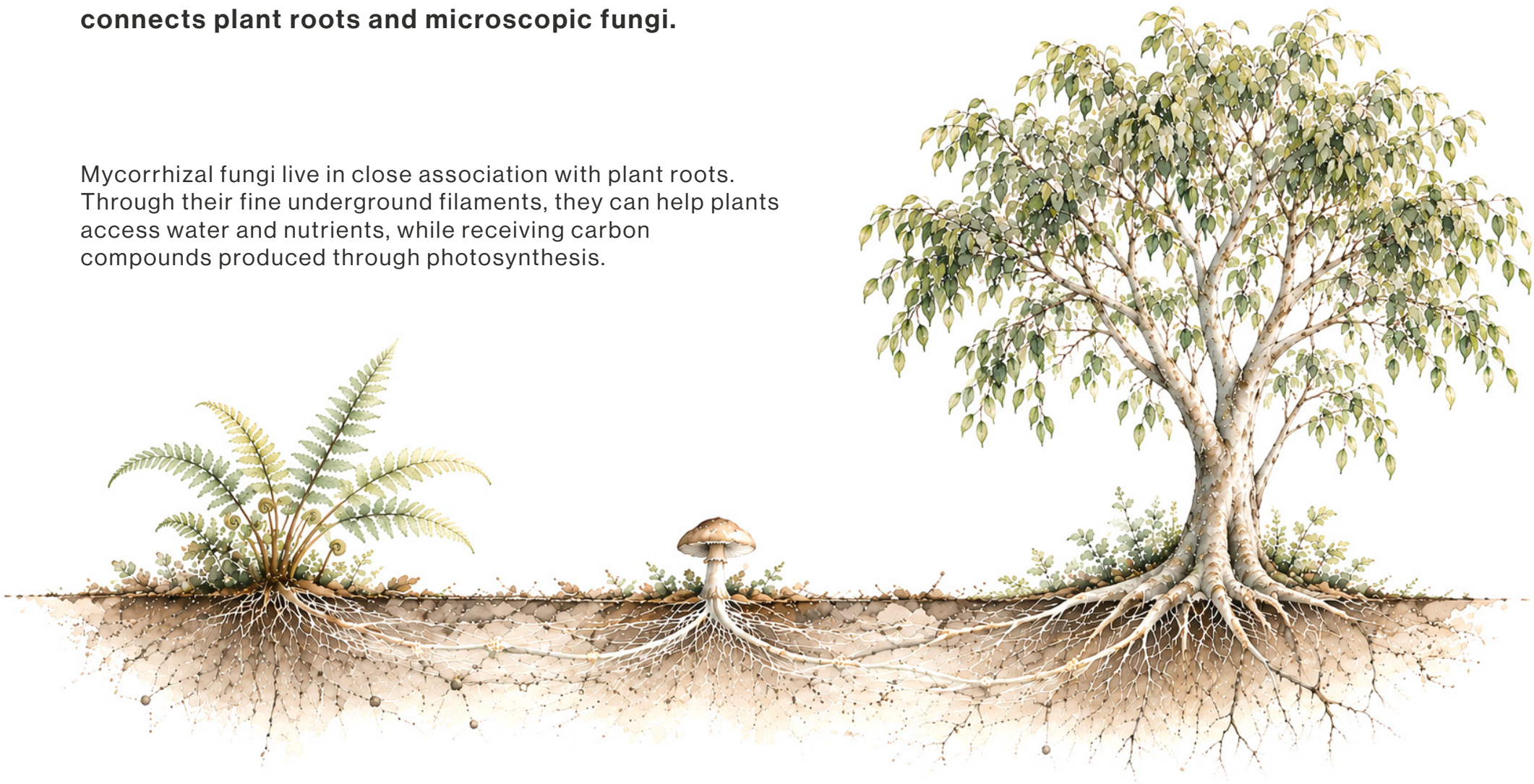


Forest types based on the Mt. Tapulao biodiversity assessment (2012). Illustration is conceptual.

Beneath every step

Beneath the forest floor, an invisible partnership connects plant roots and microscopic fungi.

Mycorrhizal fungi live in close association with plant roots. Through their fine underground filaments, they can help plants access water and nutrients, while receiving carbon compounds produced through photosynthesis.



An ancient partnership, quietly sustaining life.

An invisible question

Across five forest zones, scientists are asking a question: does the hidden diversity of *mycorrhizal fungi* change as the mountain rises?

On Mt. Tapulao, Melissa Pecundo and SPUN's *Underground Explorers* are investigating whether and how fungal communities vary with altitude, soil and vegetation, searching for patterns in a hidden world we are only beginning to understand.



Different forests. Different soils. Different fungal worlds?

Reading the soil

To discover what lives beneath the forest, scientists must learn to read the soil itself.

Even small soil samples can contain traces of fungal DNA. Through laboratory analysis and DNA sequencing, scientists can identify fungi that might otherwise remain undetected.

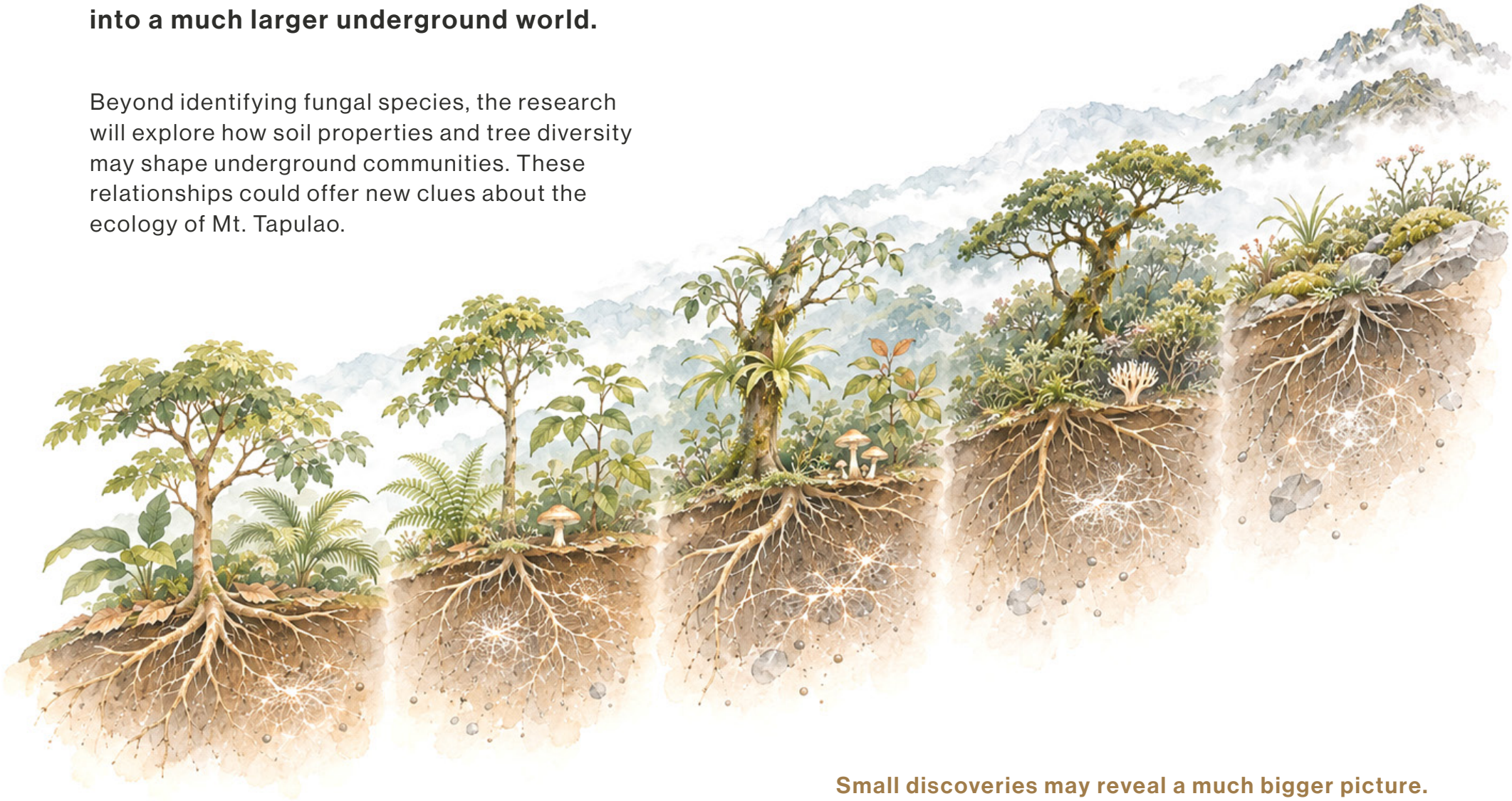


Every handful of soil holds a story we cannot see.

Patterns yet to be found

Each soil sample offers a small glimpse into a much larger underground world.

Beyond identifying fungal species, the research will explore how soil properties and tree diversity may shape underground communities. These relationships could offer new clues about the ecology of Mt. Tapulao.



Small discoveries may reveal a much bigger picture.

Why the unseen matters

The health of a forest depends on more than what grows above the ground.

Mycorrhizal fungi play an important role in forest ecosystems, supporting plant growth and nutrient cycling.

Studying their diversity can reveal more about the living systems that sustain these forests.

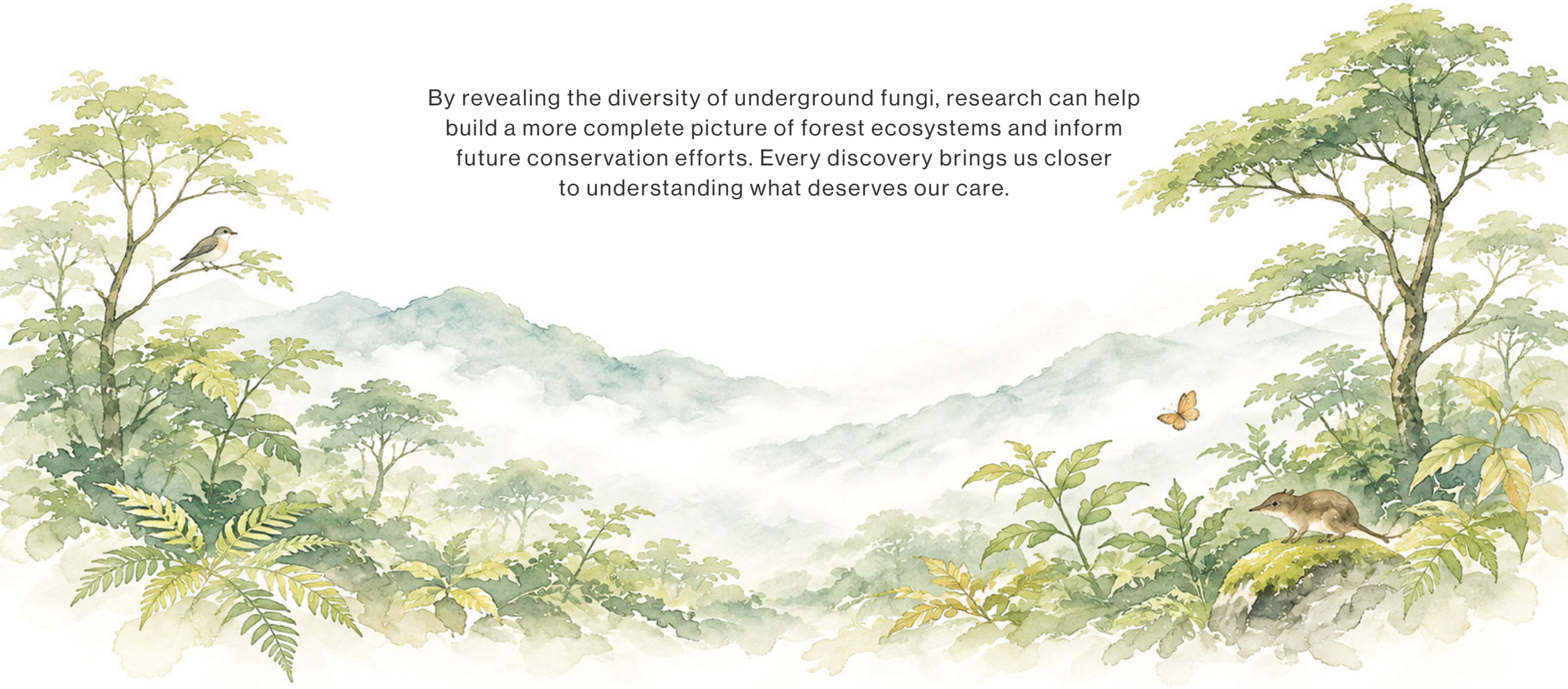


What we cannot see may be essential to what we can.

Protecting what we cannot see

To protect a forest, we must first understand the life it holds, even the life hidden beneath our feet.

By revealing the diversity of underground fungi, research can help build a more complete picture of forest ecosystems and inform future conservation efforts. Every discovery brings us closer to understanding what deserves our care.



What remains unseen should not remain unprotected.

Sources & References

SCIENTIFIC RESEARCH

01 — Balete et al. (2009)

Diversity patterns of small mammals in the Zambales Mts., Luzon, Philippines.

02 — Martin & van der Heijden (2024)

The mycorrhizal symbiosis: Research frontiers in genomics, ecology, and agricultural application.

03 — Rivera & Paz-Alberto (2022)

Assessment of the Diversity of Trees in Mt. Tapulao, Palauig, Zambales.

RESEARCH PROJECT & BACKGROUND

Pecundo / SPUN (2025)

Investigating Mycorrhizal Fungal Diversity in Zambales Mountain Range, Luzon, Philippines.

Society for the Protection of Underground Networks (SPUN)

Expedition Soil Sampling Protocol.

Philippine Information Agency (2024)

Empowering the Zambaleño Youth to Be Good Stewards of Nature.

ABOUT THIS PROJECT

An independent visual storytelling project created for educational purposes, inspired by scientific research on Mt. Tapulao. Not commissioned by or affiliated with SPUN or the researchers referenced.

Illustrations are AI-assisted artistic interpretations, not documentary records. Depicted species, ecological processes, fieldwork scenes and geographical features are conceptual and may not accurately represent actual conditions or scale.

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